

Signal Analysis & Telemetry Extraction Report

Target Recording: DEMO_SAT_QPSK_NOMINAL.iq | Format: IQ | Sample Rate: 500.0 kSps

SIGNAL HEALTH
100/100
STATUS: GOOD

ANALYST PRIORITY
LOW
Signal operating within
n...

CONFIDENCE
99.0%
DSP DATA QUALITY

ANOMALIES
0
DETECTED ISSUES

1. Explainable Analysis & Executive Summary

Automated analysis completed for 'DEMO_SAT_QPSK_NOMINAL.iq'.

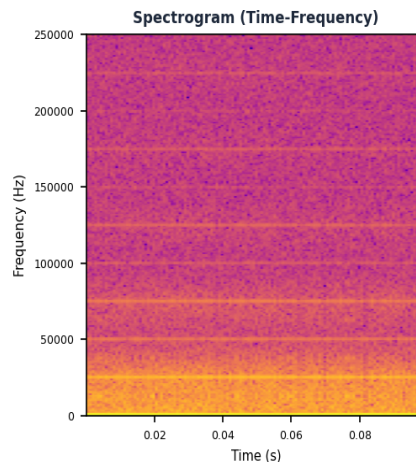
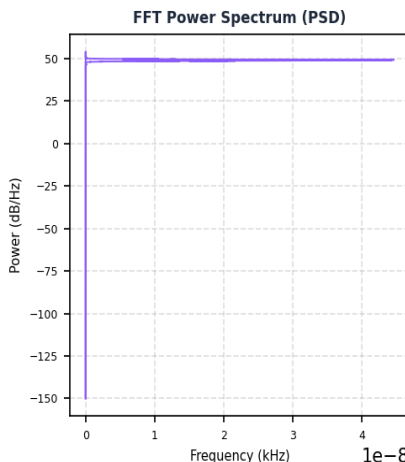
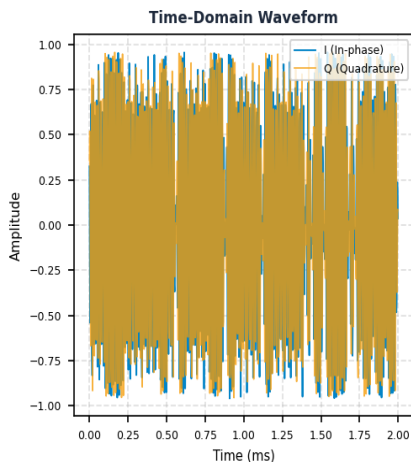
The recording contains 1 detected active segment(s) across a total duration of 1.000s. Spectral analysis identifies a center frequency at 79.997 kHz with an occupied bandwidth of 32.959 kHz and an SNR of 48.0 dB (Modulation profile: Frequency Shift Keying (FSK / 2-FSK/4-FSK)).

Benchmarked against historical baseline 'SAT-QPSK-REF-BASELINE'. Statistical drift analysis indicates baseline center frequency was 80037.0 Hz (observed: 79996.6 Hz) and nominal SNR was 48.2 dB.

No statistical anomalies detected; signal parameters conform to standard operational boundaries.

Overall Signal Health Score is 100.0/100 (GOOD), assigning this file to LOW PRIORITY for engineer triage.

2. Time-Domain, Spectral & Time-Frequency Visualization



3. Extracted Signal Parameters

MEASURED PARAMETER	VALUE	DERIVED / ESTIMATED PARAMETER	VALUE
Peak Frequency	79.834 kHz	Occupied Bandwidth (99% OBW)	32.959 kHz
Spectral Centroid	79.997 kHz	-3 dB Bandwidth	15.137 kHz
Signal Power (dBFS)	-2.26 dB	-10 dB Bandwidth	27.832 kHz
Est. Noise Floor	-92.24 dB	Signal Stability Index	0.914 / 1.0
Measured SNR	48.04 dB	Crest Factor / PAPR	1.297 (2.26 dB)
Peak Amplitude	1.0 V	Spectral Flatness (Entropy)	0.0021

RMS Amplitude	0.7712 V	Modulation Classifier Hint	Frequency Shift Keying (FSK / 2-FSK/4-FSK) (88.0%)
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4. Signal Fingerprint & 'What Changed?' Analysis (Fingerprint ID: FP-B4CA16ABC984FA89)

PARAMETER	BASELINE MEAN	CURRENT VALUE	DELTA (%)	DIR	STATUS
Center Frequency	80036.97 Hz	79996.58 Hz	-0.1%	→	Nominal
Bandwidth (99% OBW)	32958.98 Hz	32958.98 Hz	+0.0%	→	Nominal
Signal-to-Noise Ratio (SNR)	48.22 dB	48.04 dB	-0.2%	→	Nominal
Signal Power	-2.27 dBFS	-2.26 dBFS	+0.0%	→	Nominal
Signal Stability	0.913 index	0.914 index	+0.1%	→	Nominal
RMS Amplitude	0.77 V	0.771 V	+0.2%	→	Nominal

5. Anomaly Investigation Log (0 events)

No anomalies detected. All measured parameters fall within nominal mission thresholds.